MTH129 Introduction to AI for Mathematical Sciences (MS)

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SMP AI Team

Xi'an Jiaotong-Liverpool University
西交利物浦大学

Maximum Likelihood and Least Square

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Univariate Gaussian Distribution

Probability Density Function (PDF)

For a random variable X with parameter θ:

$$p(x|\theta) : \text{Probability density at } x \text{ given } \theta$$

Satisfies $\int p(x|\theta)dx = 1$ for continuous distributions.

PDF of Normal(Gaussian) Distribution

$$p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Properties: Symmetry, mean μ, variance σ^2
- Why important? Central Limit Theorem
- Connection to linear regression: Noise assumption

Likelihood Function: Formal Definition

Likelihood Function

Given observed data $\mathcal{D} = \{x_1, x_2, \ldots, x_n\}$:

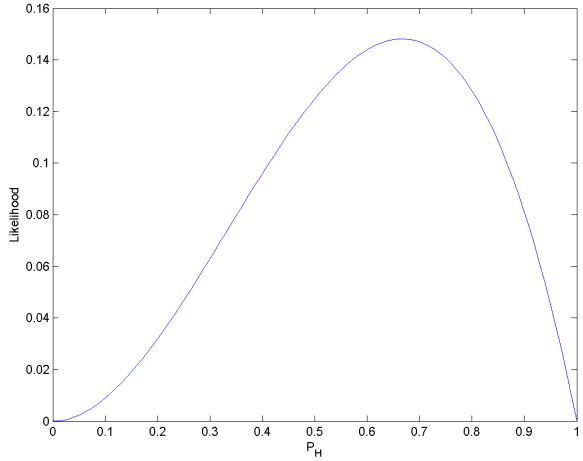
$$\mathcal{L}(\theta|\mathcal{D}) = \prod_{i=1}^{n} p(x_i|\theta)$$

- Function of θ (data fixed)
- Not a probability density (doesn't integrate to 1)
- Measures "plausibility" of θ given data

Maximum Likelihood Estimation (MLE)

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} p(x_i|\theta)$$

$$\ell(\theta) = \log \mathcal{L}(\theta)$$



Key Insight

MLE finds θ that makes observed data most probable

$$\hat{\theta}_{MLE} = \text{argmax}_{\theta} \mathcal{L}(\theta|\mathcal{D})$$

Example: MLE for Heights

Questions

Assume that the heights of students in XJTLU follows normal distribution $\mathcal{N}(\mu, \sigma^2)$, we want to build a model to model the distribution of heights of the group, based on a random sample of 10 students:

$$173, 185, 173, 174, 176, 180, 191, 194, 160, 165$$

Find the estimate for μ

Example: Analytical Derivation

Setting:

- Data: $X_1, X_2, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$
- σ^2 known, estimate μ

Likelihood function:

$$\mathcal{L}(\mu|x_i) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

Log-likelihood:

$$\begin{aligned}
\ell(\mu|x_i) &= \log \mathcal{L}(\mu|x_i) \\
&= \sum_{i=1}^{n} \left[-\frac{1}{2}\log(2\pi\sigma^2) - \frac{(x_i-\mu)^2}{2\sigma^2}\right] \\
&= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2
\end{aligned}$$

Maximizing the Log-Likelihood (1)

Take first derivative w.r.t μ:

$$\begin{aligned}\frac{d\ell}{d\mu} &= \frac{d}{d\mu}\left[\text{const} - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right] \\ &= -\frac{1}{2\sigma^2}\sum_{i=1}^{n}2(x_i-\mu)(-1) \\ &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)\end{aligned}$$

Maximizing the Log-Likelihood (2)

Set the first derivative to zero:

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)=0$$

$$\sum_{i=1}^{n}x_i-n\mu=0$$

$$\mu=\frac{1}{n}\sum_{i=1}^{n}x_i$$

Second derivative test: $\frac{d^2\ell}{d\mu^2}=-\frac{n}{\sigma^2}<0$, which is the maximum value.

Example: Calculation

Data: 10 measurements of heights given in the example (in cm):

$$\mathcal{D} = \{173, 185, 173, 174, 176, 180, 191, 194, 160, 165\}$$

Step-by-step calculation

Sample size $n = 10$

$$\hat{\mu}_{MLE} = \frac{S}{n} = \frac{1771}{10} = 177.1$$

Important Note

The MLE happens to be the sample mean? Not really! Actually in daily usage of sample mean, people are assuming the population is NORMAL!!!

Connecting MLE to Linear Regression

Assumptions:

- Gaussian noise assumption: $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$
- Conditional distribution: $y_i|\mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_{i\cdot}\mathbf{w}, \sigma^2)$

Likelihood function:

$$\mathcal{L}(\mathbf{w}) = p(\mathbf{y}|\mathbf{X}, \mathbf{w}, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_{i\cdot}\mathbf{w})^2}{2\sigma^2}\right)$$

Log-likelihood:

$$\begin{aligned}
\ell(\mathbf{w}) &= \log \mathcal{L}(\mathbf{w}) \\
&= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_{i\cdot}\mathbf{w})^2 \\
&= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2
\end{aligned}$$

Maximizing $\ell(\mathbf{w})$:

$$\begin{aligned}\text{argmax}_{\mathbf{w}}\ell(\mathbf{w}) &= \text{argmin}_{\mathbf{w}}\frac{1}{2\sigma^2}\|\mathbf{y}-\mathbf{X}\mathbf{w}\|^2 \\ &= \text{argmin}_{\mathbf{w}}\sum_{i=1}^{n}(y_i-\mathbf{x}_{i\cdot}\mathbf{w})^2\end{aligned}$$

Ignoring constants, the objective is:

$$J(\mathbf{w}) = \frac{1}{2}\|\mathbf{y}-\mathbf{X}\mathbf{w}\|^2$$

Fundamental Connection

Under Gaussian noise assumption:

$$\text{MLE} \equiv \text{Least Squares}$$

Optinal: Mahalanobis Distance

Definition: Distance measure accounting for covariance structure

$$d_M(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$$

Properties:

- Scale-invariant
- Accounts for feature correlations
- Unitless measure
- Defines equiprobability contours

Special Case: Euclidean Distance

When $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$:

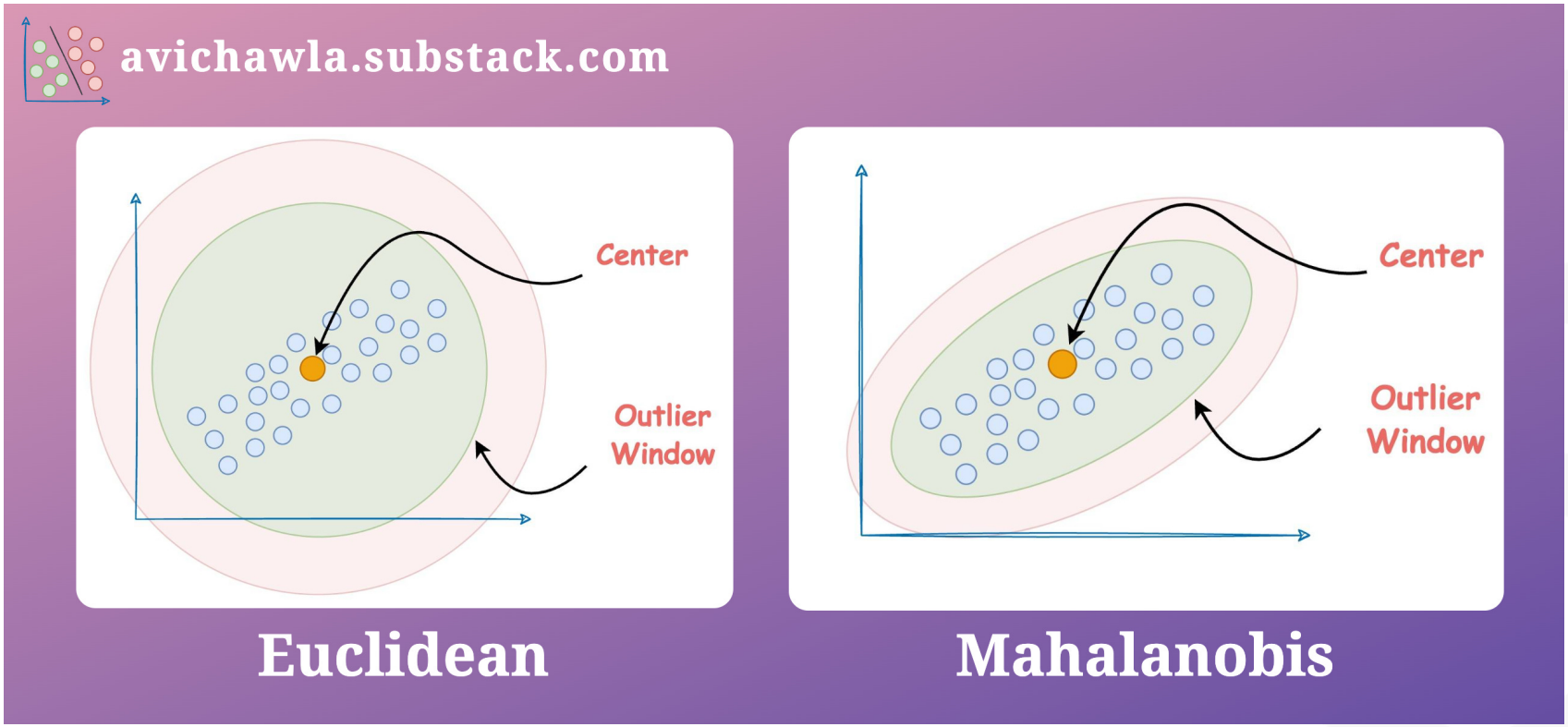
$$\begin{aligned} d_M^2(\mathbf{x}, \boldsymbol{\mu}) &= (\mathbf{x} - \boldsymbol{\mu})^T(\sigma^{-2}\mathbf{I})(\mathbf{x} - \boldsymbol{\mu}) \\ &= \frac{1}{\sigma^2}(\mathbf{x} - \boldsymbol{\mu})^T(\mathbf{x} - \boldsymbol{\mu}) \\ &= \frac{1}{\sigma^2}\|\mathbf{x} - \boldsymbol{\mu}\|^2 \end{aligned}$$

In Least Squares Context

$$\text{Mahalanobis distance} = \frac{1}{\sigma^2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2$$

- $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$ assumes:
  - Homoscedasticity (constant σ^2)
  - Uncorrelated errors ($\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$)
- This reduces to scaled Euclidean distance

Geometric Interpretation



Geometric Interpretation

General Mahalanobis:

- Sphere in transformed space: $\mathbf{z} = \mathbf{\Sigma}^{-1/2}\mathbf{x}$
- Contours are ellipsoids aligned with eigenvectors
- Distance scales with variance

Euclidean Special Case:

- $\mathbf{\Sigma} = \sigma^2\mathbf{I} \rightarrow$ isotropic scaling
- Contours are spheres
- All directions equally important

Least Squares Implication

Minimizing $\|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2$ assumes:

- Equal importance of all directions
- Uniform noise variance
- Independent measurement errors